Abdalla Tawfig

Mobile Engineer | Flutter, Kotlin

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Dubai, UAE | Arabic (native), English (fluent) | Open to remote roles

SUMMARY

Mobile Engineer with 2.5+ years building and shipping Flutter apps for iOS and Android, with native Kotlin and Swift where Flutter can't reach. Sole mobile engineer on Corteksa, a multi-tenant B2B CRM: real-time omnichannel messaging, offline-first sync, an email client and an AI assistant, across 43 releases and 4,000+ automated tests.

EXPERIENCE

Mobile Engineer **Oct 2025 - Present**

Corteksa | Saudi Arabia (Remote)

Multi-tenant B2B CRM for iOS and Android, built in Flutter with native Kotlin and Swift. · Google Play · App Store

- Built the omnichannel inbox for WhatsApp, Instagram, Messenger and TikTok over self-reconnecting WebSocket (Socket.IO) channels, with template-only replies after WhatsApp's 24-hour window and interactive buttons, lists and polls.
- Designed an offline-first message outbox on SQLite (Drift) with FIFO ordering per chat, exponential backoff, crash recovery and client-generated idempotency keys, so messages sent without signal are delivered once and never lost.
- Wrote the native push-notification layer in Kotlin and Swift behind Flutter platform channels: reply and mark-as-read even with the app killed, with an encrypted retry queue so offline replies are never dropped.
- Built deep linking with Android App Links and iOS Universal Links: shared web links open the matching screen, survive sign-in and workspace switches, fall back to the browser when needed, and are covered by contract tests.
- Built an in-app email client for Gmail, Outlook and IMAP accounts (read, reply, forward, attachments) that removes unsafe code from emails and blocks tracking images that report when a mail is opened.
- Built secure login with Google and Apple Sign-In: tokens stored encrypted on the device, automatic token refresh that keeps users signed in, and role-based permissions for what each user can see and do.
- Built the mobile client for a multi-agent AI assistant: real-time streaming of agent and tool-call events, human-in-the-loop approval before any data change, and reconnect recovery that restores pending approvals.
- Extended and own a metadata-driven UI engine that renders 33 field types across tables, Kanban boards, forms and detail views, so new business objects and fields go live through configuration, without an app release.
- Restructured the architecture of a 1,000-file Dart codebase: a shared repository guard for cache-first offline reads and consistent error handling, defensive parsing of API and socket payloads, and a unit-tested router.
- Own testing and release: 4,000+ unit, widget and BLoC tests and a CI/CD workflow that builds iOS and Android and ships to TestFlight and Google Play, across 43 releases.

Software Engineer **Jan 2024 - Jul 2025**

Weblayer | Istanbul, Turkey

- Delivered Flutter apps for B2B marketplace, CRM and employee-management clients, solo from brief to store release, with authentication, payments, real-time chat and dashboards.
- Built an internal CRM end to end with Kanban task tracking, camera-based attendance, real-time messaging and permission-scoped dashboards.
- Set Clean Architecture and BLoC as team standards, reviewed code and onboarded new engineers onto the codebase.

PROJECTS

Recallly | *Voice-to-CRM app, native Android (Kotlin, Jetpack Compose, Hilt, Room)* Play Store
Live transcription with AI extraction of CRM fields, offline speech recognition, PDF export and Google Calendar sync.

Mwaqi3 | *Arabic-first business directory for a Kuwait client* Play Store
Flutter directory app: stories, offers, Google Maps, Arabic/English RTL and a custom design system.

AutoInvo | *AI receipt scanner and expense tracker* Play Store
Camera OCR that extracts merchant, amount and date, AI categorization, offline-first sync and PDF expense reports.

TECHNICAL SKILLS

Mobile: Flutter, Dart, Kotlin (native Android), Swift (native iOS), Platform Channels, App Store Connect, Google Play Console
Architecture: Clean Architecture, BLoC/Cubit, MVVM, SOLID, Dependency Injection, Offline-first, Multi-tenant
Data & APIs: REST APIs, WebSockets (Socket.IO), OAuth 2.0, SQLite (Drift), Firebase (FCM, Crashlytics), Supabase, PostgreSQL
Quality & Delivery: Unit, Widget and BLoC Testing, Mocktail, CI/CD, Git, Claude Code, Cursor
Languages: Dart, Kotlin, Swift, Java, Python, SQL

EDUCATION

B.Sc. Software Engineering | Üsküdar University, Istanbul | GPA 3.56/4.0 **Graduated 2025**